

Design, Sampling, and Collection

Session 1

Ethan Ligon

September 3, 2026

What a careful analyst should know before touching the data.

- Deaton ch. 1 — the whole chapter
- Deaton §1.4 on survey design and §2.2 on what to do about it
- Skim the GLSS7 *Report* methodology section

Why design comes first

- A household survey is not a sample of households drawn from a hat
- It is a *designed* object: a frame, a stratification, a multistage draw
- Design determines
 - what population your estimate is an estimate of
 - how precise it is
 - which of your standard errors are lies
- None of this is recoverable after the fact from the data alone

The frame and the target population

- **Target population:** what you want to describe (all Ghanaian households, 2017)
- **Sampling frame:** the list you actually draw from (EAs from the 2010 Census)
- The gap between them is the first source of bias, and it is never zero
 - new construction, nomadic populations, institutional households
 - a frame ages: 2017 sample, 2010 frame

Multistage sampling

Two stages, in nearly every LSMS:

1. draw enumeration areas (EAs, *clusters*, PSUs) — probability proportional to size
2. within each selected EA, list the households, draw a fixed number (10–15) at random

Consequences:

- households in the same EA are *not* independent draws
- selection probability differs across households
- both must be undone, and they are undone by different machinery

Weights

If household i enters the sample with probability π_i , the Horvitz–Thompson estimator of a population total is

$$\hat{T} = \sum_{i \in S} \frac{y_i}{\pi_i} = \sum_{i \in S} w_i y_i,$$

and of a mean,

$$\hat{\mu} = \frac{\sum_{i \in S} w_i y_i}{\sum_{i \in S} w_i}.$$

- $w_i = 1/\pi_i$ is the *design weight*: household i "stands for" w_i households
- Published weights are usually design weights *adjusted* for nonresponse and calibrated to census margins

Clusters, strata, and the design effect

Ignoring clustering understates standard errors. With m households per cluster and intra-cluster correlation ρ ,

$$\text{deff} = \frac{\text{Var}_{\text{cluster}}}{\text{Var}_{\text{srs}}} \approx 1 + (m - 1)\rho.$$

- $m = 12, \rho = 0.2 \Rightarrow \text{deff} \approx 3.2$
- standard errors too small by a factor of $\sqrt{3.2} \approx 1.8$
- your 5% test is really a 20% test

Stratification cuts the other way: it *reduces* variance, so ignoring it is conservative. Ignoring clustering is not.

Three structures

Structure	Observes	Identifies
Cross-section	i once	levels, cross-sectional shape
Repeated cross-section	new i each t	aggregate change; cohorts
Panel	same i over t	within-household change

- Ghana's GLSS is (mostly) a repeated cross-section: seven rounds since 1987
- Uganda's UNPS is a genuine panel: the same households, 2005–2019
- We will use both, for exactly this reason

Attrition and the household that splits

- Attrition is *selective*: households that move, dissolve, or refuse are not a random subset
- A "household" is not a persistent object
 - a son marries and forms a new household — one household or two?
 - the head dies — is the remainder the same household?
- Panel surveys adopt a *following rule*, and it is a modelling assumption, not a clerical one
- Panel weights exist to patch this up; read how they were constructed before you use them

Exercises

1. Repeat the weighted/unweighted comparison for mean household size. Does the gap have the same sign as it did for the rural share? Explain why.
2. Compute defl for household size in 2016–17, and back out the implied ρ . How many households would a simple random sample need in order to match the precision of this survey?
3. The weights sum to the sample size, so they cannot give you a population total. Find the expansion factors in the GLSS documentation and work out what constant each round's weights were divided by. Which quantities computed above would change if you used the raw factors instead, and which are invariant?